

3 hours on examples

Problem 1 (96IRNTST)

We claim the answer is $f \equiv 0$ or $f(x) = x^2$, which obviously work.

We plug in $y = \frac{f(x)-x^2}{2}$. We may get,

$$0 = 2f(x)(f(x) - x^2)$$

Therefore, for any x , either $f(x) \in \{0, x^2\}$.

Case 1: $f(x) \neq 0$ for any $x \neq 0$. This is the $f(x) = x^2$ case.

Case 2: $f(x) = 0$ for some $x \neq 0$.

We plug in such an x to get, for all y ,

$$f(x^2 + y) = f(-y)$$

If $f(-y) = y^2$, it is impossible for $f(x^2 + y) \in \{x^4 + 2x^2y + y^2, 0\}$ to be equivalent to it unless if $x = 0$, which is impossible. Therefore, $f \equiv 0$ in this case.

Problem 3 (08IMO4)

We claim the answer is $f(x) = x$ or $f(x) = \frac{1}{x}$. The latter obviously works. The former works since $wx = yz$.

Plug in $w = x = y = z = 1$,

$$\frac{2f(1)^2}{2f(1)} = 1 \Leftrightarrow f(1) = 1$$

We next plug in $y = w; z = x$ to get, for any positive w, x ,

$$\frac{f(w)^2 + f(x)^2}{f(w^2) + f(x^2)} = 1 \Leftrightarrow f(w)^2 - f(w^2) = f(x^2) - f(x)^2$$

Plugging in $x = 1$, we get $f(w)^2 = f(w^2)$. Substituting this back in, for any positive $wx = yz$,

$$\frac{f(w^2) + f(x^2)}{f(y^2) + f(z^2)} = \frac{w^2 + x^2}{y^2 + z^2}$$

Since f is from positive reals to positive, we plug in positive numbers $a = w^2; b = x^2; c = y^2; d = z^2$ where $ab = cd$,

$$\frac{f(a) + f(b)}{f(c) + f(d)} = \frac{a + b}{c + d}$$

We now plug in $b = \frac{1}{a}; c = d = 1$.

$$f(a) + f\left(\frac{1}{a}\right) = a + \frac{1}{a} \quad (*)$$

$$\Leftrightarrow f(a^2) + 2f(a)f\left(\frac{1}{a}\right) + f\left(\frac{1}{a}\right)^2 = a^2 + 2 + \frac{1}{a^2}$$

From the other direction, we plug in $f(y)^2 = f(y^2)$; $f(z)^2 = f(z^2)$. Therefore, for any positive $wx = yz$,

$$\frac{f(w)^2 + f(x)^2}{f(y)^2 + f(z)^2} = \frac{w^2 + x^2}{y^2 + z^2}$$

We plug in $y = z = 1$ and $x = \frac{1}{w}$.

$$f(w)^2 + f\left(\frac{1}{w}\right)^2 = w^2 + \frac{1}{w^2}$$

Combining this with the previous conclusion,

$$f(a)f\left(\frac{1}{a}\right) = 1$$

Plugging this back into (*),

$$f(a) + \frac{1}{f(a)} = a + \frac{1}{a}$$

By bounding, this is possible only at $f(a) = a$ or $f(a) = \frac{1}{a}$, so $f(a) \in \left\{a, \frac{1}{a}\right\}$.

If $f(x) = x$ for some $x > 1$, then since f is continuous, $f(x) = x$ for all $x > 1$. We'd also get $f\left(\frac{1}{x}\right) = \frac{1}{x}$.

Once again, since f is continuous, $f(x) = \frac{1}{x}$ for all $x < 1$ as well too, so $f(x) = x$.

Doing analogous work on $f(x) = \frac{1}{x}$, we have either $f(x) = x$ or $f(x) = \frac{1}{x}$ as desired. (the solution resolves the point-wise trap differently and I'm concerned whether this argument with continuous is valid)

Problem 4 (77IMO6)

We let $\{f(1), f(2), \dots\} = \{a_1, a_2, \dots\}$ be the multiset of outputs of f with $a_1 \leq a_2 \leq \dots$.

Claim: $a_n = f(m) \Leftrightarrow n = m$

We prove this by induction.

Base case: $n = 1$.

FTSOC, if $a_1 = f(k)$ for some $k \neq 1$, then plug in $n = k - 1 \in \mathbb{Z}_{>0}$.

$$f(k) > f(f(k - 1))$$

This contradicts to $f(k)$ being the minimum. Therefore, $a_1 = f(1)$ and $a_1 \neq f(k)$ for any other positive integer k . This proves $a_1 = f(m) \Leftrightarrow n = 1$.

Inductive step: $n = k - 1 \rightarrow n = k$

FTSOC, let $a_k = f(x)$ for some $x \neq k$. By the inductive hypothesis, $x < k$ is impossible so $x > k$. We plug $n = x - 1$ to get,

$$f(x) > f(f(x - 1))$$

However, $f(x - 1) \geq a_k$ by the induction hypothesis so $f(f(x - 1)) \geq a_k$ as well. There is hence a contradiction, and we are done.

Conclusion: By mathematical induction, the claim is proven.

Since the claim gives us injectivity as well as $f(1) \leq f(2) \leq \dots$, we have, f is strictly increasing. Therefore, $f(m) > f(n) \Leftrightarrow m > n$. Using this on our initial equation,

$$n + 1 > f(n)$$

$$\Leftrightarrow n \geq f(n)$$

Since $f: \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0}$, we quickly induct to get $f(n) = n$.

8 clubs; 5.5 hours

Problem 5 (17EGMO2)

We claim the answer is $k = 3$.

This can be achieved by coloring all the $n \equiv 0 \pmod 3$ one color, all the $n \equiv 1 \pmod 3$ another, and all the $n \equiv 2 \pmod 3$ yet another color. Our function f would be,

$$f(x) = \begin{cases} x, & \text{if } x \equiv 1, 2 \pmod 3 \\ 1434x, & \text{if } x \equiv 0 \pmod 3 \end{cases}$$

For two m, n of the same color, if $m \equiv n \equiv 1$ or $2 \pmod 3$, then $f(m) + f(n) = m + n = f(m + n)$. If $m \equiv n \equiv 0 \pmod 3$, then $f(m) + f(n) = 1434(m + n) = f(m + n)$. In this case, $m = 1; n = 2$ satisfies the second condition as $f(1 + 2) = 1434 \cdot 3 \neq 3$.

We now show the stronger claim that $k = 2$ is impossible (which shows $k = 1$ is impossible as well since you can choose one of the colors to fill everything with) even if the range of the function is all positive reals. Name the colors A, B .

Since the equalities are homogenous, we scale to let $f(1) = 1$ WLOG.

We seek to prove that if $k = 2$, $f(x) = x \forall x$.

Note that, since m has the same color as itself, $f(2m) = f(m) + f(m) = 2f(m)$. Therefore,

Lemma: $f(x) = x \Leftrightarrow f(2x) = 2x$

We now prove the claim using strong induction.

Base case: $x = 1$ is trivial.

Inductive step ($x = 1, \dots, k - 1 \rightarrow x = k$):

FTSOC, let $f(k) \neq k$.

Note that if k is even, then we've already proved $\frac{k}{2}$ so by the lemma $f(k) = k$. (!)

For odd k , we have already proved $\frac{k+1}{2}, \frac{k+3}{2}, \dots, \frac{2k-2}{2}$ so,

$$f(k+1) = k+1; f(k+3) = k+3; \dots; f(2k-2) = 2k-2$$

WLOG, let k be colored A . Note that pairs $(1, k-1), (2, k-2), \dots, (\frac{k-1}{2}, \frac{k+1}{2})$ must have different colors since otherwise we may use (i) to get $f(k) = k$. In each of these pairs, if the odd element o is colored A , we have,

$$f(o) + f(k) = f(o+k)$$

Since $o+k$ is an even number less than or equal to $2k-2$,

$$\Leftrightarrow o + f(k) = o + k \Leftrightarrow f(k) = k \quad (!)$$

Therefore, all the odd numbers less than k are colored B and all the evens are colored A . Therefore, we add to get $f(k+2) = f(k) + 2$. If $k+2$ is colored B , then,

$$f(1) + f(k+2) = f(k+3)$$

$$\Leftrightarrow f(k) + 3 = k + 3 \quad (!)$$

However, if $k+2$ is colored A ,

$$f(k) + f(k+2) = f(2k+2)$$

$$\Leftrightarrow 2f(k) + 2 = 2(k+1) \quad (!)$$

Using proof by contradiction, the inductive step is complete.

Conclusion: By strong mathematical induction, we have proved that $f(k) \equiv k$.

If $f(x) = x$, then we cannot have $f(m+n) = f(m) + f(n)$ so $k = 2$ does not work.

Problem 7 (EGMO 2021/2)

We claim that the solutions are $f(x) \equiv x$ or $f(x) \equiv -x$. These

Lemma: $a \neq 0 \Rightarrow f(a) \neq 0$

FTSOC, $a \neq 0$ and $f(a) = 0$, we plug in $x = a; y = 0$,

$$f(a f(a)) = f(0) + a^2$$

$$\Leftrightarrow f(0) = f(0) + a^2 \Leftrightarrow a = 0 \quad (!)$$

If we plug in $y = kxf(x)$ for some integer k ,

$$f((k+1)xf(x)) = f(kxf(x)) + x^2$$

Therefore, by induction, for any quotients x, y and integer n ,

$$f(n \cdot xf(x)) = f(0) + n \cdot x^2$$

Since $xf(x)$ is a quotient, for any non-zero quotients a, b , we can choose non-zero integers m, n such that,

$$\begin{aligned} m \cdot af(a) &= n \cdot bf(b) \\ \Rightarrow f(m \cdot af(a)) &= f(n \cdot bf(b)) \\ \Leftrightarrow f(0) + m \cdot a^2 &= f(0) + n \cdot b^2 \\ \Leftrightarrow ma^2 &= nb^2 \end{aligned}$$

Since m, a are non-zero, $f(a)$ is non-zero as well. $m \cdot af(a) = n \cdot bf(b) \neq 0$. We can hence divide the last equation by the first to get, for any non-zero quotients a, b .

$$\begin{aligned} \frac{ma^2}{m \cdot af(a)} &= \frac{nb^2}{m \cdot bf(b)} \\ \Leftrightarrow \frac{a}{f(a)} &= \frac{b}{f(b)} \end{aligned}$$

Fixing a non-zero b , we have $\frac{a}{f(a)} = k$ for a constant quotient k . Therefore,

$$f(x) \equiv kx$$

We plug this in,

$$f(kx^2 + y) = ky + x^2$$

Plugging in $y = 0$,

$$\Rightarrow k^2x^2 = x^2$$

Therefore, $k = \pm 1$ and we are done.

Problem 8 (11TSTST1)

We claim that either $f(a, b) \equiv a$ or $f(a, b) \equiv b$.

Plugging in $a = b = c$, the median of $f(a, a), f(a, a), f(a, a)$ is a . Therefore,

$$f(a, a) = a$$

Now plugging in $b = a \neq c$, plugging in,

$$\text{median}(f(a, b), f(b, c), f(c, a)) = \text{median}(a, b, c)$$

$$\Leftrightarrow \text{median}(f(a, a), f(a, c), f(c, a)) = a$$

Since $f(a, a) = a$ already, for any distinct integers a, c ,

$$a \in \{f(a, c), f(c, a)\}$$

By symmetry, we also have,

$$c \in \{f(a, c), f(c, a)\}$$

Therefore, we get, $f(a, b) \in \{a, b\}$ and $f(b, a)$ is the other selection.

We resolve the point-wise trap using proof by contradiction.

FTSOC, let, for numbers $a \neq b; c \neq d$,

$$f(a, b) = a; f(c, d) = d$$

If $a > b$, we may switch the positions of a, b to get $f(b, a) = b$. WLOG, let $a < b$ and $c < d$.

Claim: $f(a, b) = a \Rightarrow f(a, d) = a$ whenever $a < b, d$ or $a > b, d$.

FTSOC, if $f(a, d) = d$ in this case,

$$f(a, b) = a; f(a, d) = d \Leftrightarrow f(a, b) = a; f(d, a) = a$$

Therefore, the median of $f(a, b), f(b, d), f(d, a)$ is a . However, we are given that the median of a, b, d is not a . (!)

We immediately have $a = c$ is impossible. WLOG, let $a < c$. Since $d > c$ and $d > c > a$,

$$f(c, d) = d \Leftrightarrow f(d, c) = c \Rightarrow f(d, a) = c$$

Since $c \neq a; c \neq d$, we have a contradiction, and the point-wise trap is resolved.

17 clubs; 9.5 hours

Problem 10 (18TSTST1)

We claim the solution is $\theta(p) \equiv p(n)$ for any integer n . This clearly works as $p(n) \mid p(n)q(n)$.

For $p(x) = x$, we let $\theta(p) = n$ for some integer n . We prove that $\theta(p) = p(n)$ for all polynomials p using induction on the degree of p .

Base case: $\deg(p) = 0$

We plug in $p(x) = a$.

$$\theta(p + 1) = \theta(p) + 1 \Leftrightarrow \theta(a + 1) = \theta(a) + 1$$

By induction, we may write, for any integer a , $\theta(a) \equiv a + k$ for some constant k . Plugging in $p(x) = a$ and $q(x) = 0$, $a + k \mid k$. This is true for all integers a , so anything divides k meaning $k = 0$. For degree zero $p(x) = a$, $\theta(p) = a$ as desired.

Base case 2: $\deg(p) = 1$

We have $\theta(x + b) = n + b$ for any integer b . Therefore, for all integers $a, k \neq -c$,

$$\begin{aligned}\theta(x + k) &\mid \theta(ax + ak) \\ \Rightarrow n + k &\mid \theta(ax) + ak \\ \Leftrightarrow n + k &\mid \theta(ax) - an\end{aligned}$$

For a sufficiently large k , we have $\theta(an) = an$ for the RHS to be zero. Therefore, $\theta(ax + b) = an + b$.

Strong inductive step: $\deg(p) = k$ for $k \geq 2$

For any polynomial p and integers a, b , we have $a - b \mid p(a) - p(b)$.

For any integer m for $m \neq -n$, note that,

$$\begin{aligned}\theta\left(\frac{p - p(m)}{x - m}\right) &\mid \theta(p - p(n)) \\ \Leftrightarrow \frac{p(n) - p(m)}{n - m} &\mid \theta(p) - p(n) \\ \Leftrightarrow \frac{p(n) - p(m)}{n - m} &\mid \theta(p) - p(m)\end{aligned}$$

As $\deg(p) \geq 2$, as $n \rightarrow \infty$, the LHS grows increasingly greater while the RHS remains fixed. Therefore,

$$\theta(p) = p(m)$$

Problem 12 (07MOPFE)

We claim the answer is $f(x) = \frac{1}{x}$, which obviously works.

We show that for positive real numbers x, y .

$$\begin{aligned}f(x)y + \frac{x}{y} &= x^2 + y^2 \\ \Leftrightarrow 0 &= y^3 - f(x)y^2 + x^2y - x\end{aligned}$$

We let $g(y) = y^3 - f(x)y^2 + x^2y - x$ and show that g has a real positive root. Since g is continuous $\lim_{y \rightarrow 0^+} g(y) = -x$ and $\lim_{y \rightarrow +\infty} g(y) = +\infty$ noting that x is non-zero so $-x$ is negative, by the Intermediate Value Theorem, there must be a real positive root. (The IVT is usually applied on closed intervals so idk if this is valid)

We may plug such a y in to get, for some $t = f(x)y + \frac{x}{y} = x^2 + y^2$,

$$f(t) = xy \cdot f(t)$$

Since f is positive,

$$\Leftrightarrow 1 = xy$$

Plugging $y = \frac{1}{x}$ back into, $y^3 - f(x)y^2 + x^2y - x$,

$$0 = \frac{1}{x^3} - \frac{f(x)}{x^2} + x - x$$

$$\Leftrightarrow 0 = 1 - x f(x)$$

$$\Leftrightarrow f(x) = \frac{1}{x}$$

This is as desired.

25 clubs; 12.5 hours

Problem 15 (21CAMO2)

We claim that the answer is, for any set S closed under multiplication and division,

$$f(x) = \begin{cases} 0 & \text{if } x \notin S \\ \sqrt{x} & \text{if } x \in S \end{cases}$$

We may verify that this works. For any pair x, y ,

Case 1: $f(x) = 0; f(y) = 0$. Trivially both sides are zero.

Case 2: $f(x) = 0; f(y) = \sqrt{y}$

Plugging in, we indeed have,

$$(\sqrt{y})^3 = y\sqrt{y} + 3f(xy)(0 + \sqrt{y})$$

However, since S is division, if $f(xy) = \sqrt{xy}$ and $f(y) = \sqrt{y}$ then $f(x) = \sqrt{x}$, which is false. Therefore, $f(xy) = 0$ and this is clearly true.

The case for $f(x) = \sqrt{x}; f(y) = 0$ is analogous.

Case 3: $f(x) = \sqrt{x}; f(y) = \sqrt{y}$

We have $f(xy) = \sqrt{xy}$. We plug in to get this is true.

We plug in $x = y = 1$ to get,

$$8f(1)^3 = 2f(1) + 6f(1)^2$$

The non-negative solutions to this are $f(1) = 0$ and $f(1) = 1$.

Case 1: $f(1) = 0$

We claim this implies $f \equiv 0$. FTSOC, if there exists a y that gives $f(y) \neq 0$,

We plug in $x = 1$ to get,

$$\begin{aligned} f(y)^3 &= y f(y) + 3 f(y)^2 \\ \Leftrightarrow f(y)^2 - 3 f(y) - y &= 0 \end{aligned}$$

Since $y > 0$, the only positive solution for $f(y) = \frac{3+\sqrt{9+4y}}{2}$ by the quadratic formula. However, plugging in $x = y$ into the original equation for this y ,

$$(3 + \sqrt{9 + 4x})^3 = x(3 + \sqrt{9 + 4x}) + 3f(x^2)(3 + \sqrt{9 + 4x})$$

Obviously, if $f(x^2) = 0$, then by bounding this is impossible. Therefore, $f(x^2) \neq 0 \Rightarrow f(x^2) = \frac{3+\sqrt{9+4x^2}}{2}$. Plugging in,

$$\begin{aligned} (3 + \sqrt{9 + 4x})^3 &= x(3 + \sqrt{9 + 4x}) + 3 \frac{(3 + \sqrt{9 + 4x^2})}{2} (3 + \sqrt{9 + 4x}) \\ \Leftrightarrow (3 + \sqrt{9 + 4x})^2 &= x + 3 \frac{(3 + \sqrt{9 + 4x^2})}{2} \end{aligned}$$

Once again, by bounding, there are no positive solutions.

This gives $f \equiv 0$ one of our solutions.

Case 2: $f(1) = 1$

We plug in $x = 1$ to get,

$$\begin{aligned} (1 + f(y))^3 &= 1 + y f(y) + 3 f(y)(1 + f(y)) \\ \Leftrightarrow f(y)^3 + 3 f(y)^2 + 3 f(y) &= y f(y) + 3 f(y) + 3 f(y)^2 \\ \Leftrightarrow f(y)^3 &= y f(y) \end{aligned}$$

Therefore, the non-negative solutions are either $f(y) = 0$ or $f(y) = \sqrt{y}$.

However, if $f(x) = \sqrt{x}$, $f(y) = \sqrt{y}$, we have, plugging in,

$$\begin{aligned} (\sqrt{x} + \sqrt{y})^3 + x\sqrt{x} + y\sqrt{y} + 3f(xy)(\sqrt{x} + \sqrt{y}) \\ \Leftrightarrow f(xy) = \sqrt{xy} \end{aligned}$$

Therefore, S is a multiplicative subgroup.

Problem 16 (18TURTST2)

We claim that $f(x) \equiv 2x$, which obviously works.

We plug in $x = f(y) - 2y$ to get, for all real y ,

$$\begin{aligned} f(f^2(y) - 2yf(y) + y^2) &= f((f(y) - y)^2) - (f(y) - 2y)f(f(y) - 2y) \\ &\Leftrightarrow (f(y) - 2y) \cdot f(f(y) - 2y) = 0 \quad (*) \end{aligned}$$

Taking $y = 0$, we see that either $f(0) = 0$ or $f(f(0)) = 0$. However, $f(0) = 0$ implies $f(f(0)) = 0$ anyways so $f(f(0)) = 0$ is always true.

Claim: $f(a) = 0 \Rightarrow a = f(0)$.

FTSOC, if $a \neq 0$ and $f(a) = 0$, we plug in $(x, y) = (a, f(0))$ to get,

$$f(f(0)^2) = f((a + f(0))^2)$$

Plugging in $(x, y) = (f(0), a)$,

$$f(a^2) = f((f(0) + a)^2)$$

Clearly,

$$f(f(0)^2) = f(a^2)$$

We now plug in $y = a$ into the original equation,

$$f(a^2) = f((x + a)^2) - x f(x)$$

Similarly, plugging in $y = f(0)$ gives,

$$f(f(0)^2) = f((x + f(0))^2) - x f(x)$$

This means $f((x + a)^2) = f((x + f(0))^2)$.

Next, we plug in, $(x, y) = (x + a - f(0), f(0))$,

$$\begin{aligned} f(f(0)^2) &= f((x + a)^2) - (x + a - f(0))f(x + a - f(0)) \\ &\Leftrightarrow f((x + f(0))^2) - x f(x) = f((x + f(0))^2) - (x + a - f(0))f(x + a - f(0)) \end{aligned}$$

We let $c := a - f(0)$,

$$x f(x) = (x + c) f(x + c)$$

We now plug in $(x, y) = (x + c, y)$ into the original equation,

$$f((x + c) f(y) + y^2) = f((x + y + c)^2) - (x + c) f(x + c)$$

Combining this with the first equation, since $f((x + y + c)^2) = f((x + y)^2)$,

$$f((x + c) f(y) + y^2) = f(x f(y) + y^2)$$

As f is surjective, we plug in the selection $y = k$ such that $f(y) = 1$,

$$f(x + c + k^2) = f(x + k^2)$$

Since x can be any real number,

$$\Leftrightarrow f(x + c) = f(x)$$

However, using $(x + c) f(x + c) = x f(x)$, we have $x + c = x \Leftrightarrow c = 0$. (!) Therefore we are done.

Claim: $f(0) = 0$

Plugging in $y = f(0)$ into (*),

$$(f(f(0)) - 2f(0)) \cdot f(f(f(0)) - 2f(0)) = 0$$

FTSOC, if $f(0) \neq 0$, then the first factor is non-zero and,

$$f(-2f(0)) = 0$$

$$\Rightarrow -2f(0) = f(0)$$

Therefore, $f(0) = 0$. (!)

Claim: $f(x) = 2x$

We simply use (*), FTSOC, if there exists a quantity a such that $f(a) \neq 2a$,

$$(f(a) - 2a) \cdot f(f(a) - 2a) = 0$$

$$\Leftrightarrow f(f(a) - 2a) = 0$$

$$\Leftrightarrow f(a) = 2a \quad (!)$$

We have proved what was necessary.

33 clubs; 14 hours

Problem 18 (12AMO4)

We claim the solutions are $f \equiv 1, f \equiv 2, f(x) \equiv x$. These obviously work.

Claim: $f(n) \geq n$ or $f(n) \in \{1, 2\}$

FTSOC, if not, plug in $m = n!$ to get,

$$n! - n \mid f(n!) - f(n) \Leftrightarrow n! - n \mid f(n)! - f(n)$$

By bounding, this is clearly impossible for $n > 2$ and $f(n) > 2$.

Case 1: $f(n) = 1$ for some $n > 2$

We claim that $f \equiv 1$. FTSOC, if there exists $f(m) = k \neq 1$, plug in m to get,

$$m - n \mid k - f(n) \quad \forall n \neq m$$

Note that if $f(n) = 1$, we have $f(n!) = f(n)! = 1$ and so on. Therefore, we may plug in a sufficiently large value of n on the LHS and still have $f(n) = 1$. Therefore, $k = 1$.

Case 2: $f(n) = 2$ for some $n > 2$

This case is analogous, we have $f \equiv 2$.

Case 3: $f(n) \geq n$ for some $n > 2$

We use induction to prove that $f(x) \equiv x$.

Claim: $f(2) = 2$

Note that $f(2) \in \{1, 2\}$. FTSOC, let $f(2) = 1$, we have, for any $m > 2$,

$$m! - 2 \mid f(m)! - 1$$

Since $f(m) \geq m$, the RHS is odd, but the LHS is even. (!)

Claim: $f(1) = 1$

FTSOC, if $f(1) = 2$, we plug in $m = 3!$; $n = 1$ into the original equation to get,

$$3! - 1 \mid f(3)! - 2$$

Since $f(3)! - 2 \equiv 0 \pmod{5}$, clearly $f(3) \geq 5$ doesn't work. We may check to find $f(3) = 2$ is the only solution. (!)

Claim: $f(3) = 3$

We plug in $m = 3!$; $n = 1$ into the original equation to now get,

$$3! - 1 \mid f(3)! - 1$$

By a similar process, we may deduct $f(3) = 3$.

We now finally prove that $f(x) \equiv x$. FTSOC, if there exists a $n > 3$ such that $f(n) \neq n$, we plug in such a n to have,

$$m - n \mid f(m) - f(n) \quad \forall m \neq n$$

Note that, $f(3!) = 3!$; $f((3!)!) = (3!)!$ and so on. We plug in a sufficiently large m in the pattern to get, by bounding, $f(n) = n$.

Problem 20 (13IMO5)

Claim 1: $f(x) \geq x$

We let $g(a) = f(a) - a$,

$$f(x)f(y) \geq f(xy) \Leftrightarrow xg(y) + yg(x) \geq 0;$$

There exists a rational $a > 1$ with $f(a) = a \Leftrightarrow$ There exists a rational $a > 1$ with $g(a) = 0$.

We plug in $x = a$ to get, for all y in the domain of f ,

$$a g(y) \geq 0 \Leftrightarrow g(y) \geq 0 \Leftrightarrow f(y) \geq y$$

Claim 2: If $f(t) = t$, then $f(r) = r$ for all $r \leq t$.

Note that,

$$f(x + y) \geq f(x) + f(y) \Leftrightarrow g(x + y) \geq g(x) + g(y);$$

Plugging in $x = r; y = t - r$ we are done using claim 1.

Applying (i) repeatedly, we have, using claim 1,

$$f(a)^k = a^k \geq f(a^k) \Rightarrow f(a^k) = a^k \quad \forall \text{ integer } k$$

Therefore, using claim 2, as $a > 1$, we may get $f(x) \equiv x$ as desired.

Problem 21 (16SLA4)

Claim: $f(1) = 1$

Plug in $x = y = 1$,

$$\begin{aligned} f(1)f(f(1)) + f(f(1)) &= f(1)f(f(1)) + f(1)f(f(1)) \\ \Leftrightarrow f(f(1)) &= f(1)f(f(1)) \end{aligned}$$

Since f is non-zero, we are done. Plugging in $x = 1$,

$$\begin{aligned} f(f(y)) + f(y) &= f(y)(1 + f(f(y^2))) \\ \Leftrightarrow f(f(y)) &= f(y) \cdot f(f(y^2)) \quad (A) \end{aligned}$$

However, plugging in $y = 1$,

$$\begin{aligned} x f(x^2) + f(f(x)) &= f(x)(f(f(x^2)) + 1) \\ \Leftrightarrow x f(x^2) + f(f(x)) &= f(f(x)) + f(x) \\ \Leftrightarrow x f(x^2) &= f(x) \quad (B) \end{aligned}$$

Claim: f is injective

FTSOC, if $f(a) = f(b)$ but $a \neq b$,

For any positive real c , we plug in $(x, y) = (a, c)$,

$$\begin{aligned} a f(a^2) f(f(c)) + f(c f(a)) &= f(ac) (f(f(a^2)) + f(f(c^2))) \\ \Leftrightarrow f(a) f(f(c)) + f(c f(a)) &= f(ac) \left(\frac{f(f(a))}{f(a)} + \frac{f(f(c))}{f(c)} \right) \end{aligned}$$

Plugging in $(x, y) = (b, c)$,

$$f(b) f(f(c)) + f(c f(b)) = f(bc) \left(\frac{f(f(b))}{f(b)} + \frac{f(f(c))}{f(c)} \right)$$

Substituting $f(a) = f(b) \Rightarrow f(f(a)) = f(f(b))$ and dividing,

$$f(ac) = f(bc) \quad \forall c \in (0, \infty)$$

We may plug in $c = f(a) = f(b)$ to get,

$$f(a f(a)) = f(b f(b))$$

Plugging in $(x, y) = (a, a)$, using (A) and (B).

$$\Leftrightarrow f(a) f(f(a)) + f(a f(a)) = 2f(a^2) f(f(a^2))$$

$$\Leftrightarrow af(a)f(f(a)) + af(af(a)) = 2f(a)f(f(a^2))$$

$$\Leftrightarrow a(f(a)f(f(a)) + f(af(a))) = 2f(f(a))$$

Applying the same for b , we see that $f(a)f(f(a)) + f(af(a)) = f(b)f(f(b)) + f(bf(b))$ and $2f(f(a)) = 2f(f(b))$. Therefore, $a = b$.

Claim: $f(x) = \frac{1}{x}$.

Note that, by (B),

$$f(x) f(f(x)^2) = f(f(x)) \Leftrightarrow f(f(x)^2) = \frac{f(f(x))}{f(x)}$$

Using (A), however,

$$\frac{f(f(x))}{f(x)} = f(f(x^2))$$

Therefore,

$$f(f(x)^2) = f(f(x^2))$$

$$\Leftrightarrow f(x)^2 = f(x^2)$$

Applying (B) once more,

$$\Leftrightarrow f(x)^2 = \frac{f(x)}{x}$$

$$\Leftrightarrow f(x) = \frac{1}{x}$$